SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION with OpenCV   COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

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1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Compute the transformed values.r = [50, 100,s150= ,255200]− r

(c) Apply a negative transformation to the pixel values:

ANS: **Grey Level Transformation**

(a) Purpose

Grey level transformation is used to improve the quality of an image by changing the pixel intensity values. It helps to enhance brightness, contrast, and image details.

(b) Compute the transformed values

Given:

r = [50, 100, 150, 200]

Formula:

$s = 255 - r$

- $255 - 50 =$ **205**
- $255 - 100 =$ **155**
- $255 - 150 =$ **105**
- $255 - 200 =$ **55**

Answer: s = [205, 155, 105, 55]

(c) Negative Transformation

Formula:

$s = 255 - r$

Output:

[205, 155, 105, 55]

Result: Negative transformation changes bright pixels to dark and dark pixels to bright.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels. **(b)** Given s = $clog$ (r1=+[0r), ,10 where, 50, 100 c =] 10, compute s for:

ANS: **Log Transformation**

(a) Explain how log transformation enhances low-intensity pixels.

Log transformation increases the brightness of dark (low-intensity) pixels and compresses bright pixels. It helps to reveal details in dark regions of an image.

(b) Compute (s) for (r = [0, 10, 50, 100])

Given:

- **(c = 10)**

- **Formula:**

  [

  s = c \log(1 + r)

  ]

Using log base 10:

- (r = 0:; s = 10\log(1) = 0)
- (r = 10:; s = 10\log(11) \approx 10.41)
- (r = 50:; s = 10\log(51) \approx 17.08)
- (r = 100:; s = 10\log(101) \approx 20.04)

Answer:

r	s = 10 log(1+r)
0	0.00
10	10.41
50	17.08

r	s = 10 log(1+r)
100	20.04

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

ANS: **3. Histogram Processing**

(a) Role of Histogram Equalization

Histogram equalization is used to improve the contrast of an image by spreading the pixel intensity values over the full intensity range. It makes dark areas brighter and bright areas clearer.

(b) Compute the Cumulative Distribution and Equalized Intensity Values

Given (3-bit image, (L = 8)):

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels = 2 + 2 + 4 + 8 = 16

Intensity	Frequency	Cumulative Frequency	Equalized Value ((L-1)\times \frac{CF}{16})
0	2	2	**1**
1	2	4	**2**
2	4	8	**4**
3	8	16	**7**

Answer:

- **Cumulative Distribution (CF): 2, 4, 8, 16**
- **Equalized Intensity Values: 1, 2, 4, 7**

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region: Compute the new center

pixel[10 value. 20 30 20 30 40 30 40 50]

ANS: Got it. I'll avoid tables, equations, and formatting so you can

copy-paste directly into MS Word.

4. Spatial Filtering (Smoothing)

(a) Purpose:

Spatial filtering is used to reduce noise and smooth an image by replacing each pixel

with the average of its neighboring pixels. This improves image quality.

(b) Calculation:

Given region:

10 20 30

20 30 40

30 40 50

Sum = 10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270

Average = 270 / 9 = 30

Answer: New center pixel value = **30**

From now on, I'll give all your **Computer Vision** answers in this plain text format so

they paste perfectly into Word.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

to the matrix: $[[0 - 1 \quad 0\}, \{- 1 \quad 4 - 1], \{ 0 - 1 \quad 0]$]

Compute the output at the[10 center. 10 10 , 10 20 10, 10 10 10]

ANS: **5. Spatial Filtering (Sharpening)**

(a) Purpose:
Sharpening enhances image details by highlighting edges and fine features. It increases the contrast between neighboring pixels, making the image appear clearer.

(b) Calculation:

Laplacian mask:

0 -1 0

-1 4 -1

0 -1 0

Image:

10 10 10

10 20 10

10 10 10

Output = (4 × 20) − (10 + 10 + 10 + 10)

= 80 − 40

= 40

Answer: Output at the center = **40**

6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain. **(b)** Given frequency components:

Apply a low-pass filter that retains$F$ only= [100 the, 80first, 40 two, 20 components.] Write the filtered output.

ANS: **6. Frequency Domain Filtering**

(a) Difference between Low-Pass and High-Pass Filtering:

- **Low-pass filter:** Passes low-frequency components and removes high-frequency components. It is used for smoothing and noise reduction.
- **High-pass filter:** Passes high-frequency components and removes low-frequency components. It is used for edge detection and image sharpening.

(b) Calculation:

Given:

F = [100, 80, 40, 20]

Low-pass filter retains the first two components.

Filtered output:

F = [100, 80, 0, 0]

Answer: [100, 80, 0, 0]

7. Thresholding

(a) Explain global thresholding in segmentation. **(b)** Given pixel values:

Apply threshold T = 100. Assign values[200(0 or] 1).

ANS: **7. Thresholding**

(a) Global Thresholding:
Global thresholding is an image segmentation technique that uses a single threshold value to separate the foreground from the background. Pixels greater than or equal to the threshold are assigned **1**, and pixels below the threshold are assigned **0**.

(b) Calculation:

Given pixel values:

[50, 100, 150, 200]

Threshold:

T = 100

Applying the threshold:

- $50 \rightarrow 0$
- $100 \rightarrow 1$
- $150 \rightarrow 1$
- $200 \rightarrow 1$

Answer: [0, 1, 1, 1]

8. Edge Detection

(a) Explain the concept of edge detection.

(b) Apply the Sobel operator (horizontal):

to: $[-1 -2 -1, 0 \quad 0 \quad 0, 1\ 2\ 1\ ]$

Compute the gradient at[10the 10center. 10, 20 20 20, 30 30 30]

ANS: **Edge Detection**

(a) Edge Detection:
Edge detection is used to identify the boundaries of objects in an image by detecting sudden changes in pixel intensity.

(b) Calculation:

Sobel (Horizontal) mask:

-1 -2 -1

0 0 0

1 2 1

Image:

10 10 10

20 20 20

30 30 30

Gradient = (-1×10) + (-2×10) + (-1×10) + (0×20) + (0×20) + (0×20) + (1×30) + (2×30) + (1×30)

= -10 -20 -10 + 0 + 0 + 0 + 30 + 60 + 30

= **80**

Answer: Gradient at the center = **80**

9. Region-Based Segmentation

(a) Explain region growing in segmentation. **(b)**
Given pixel intensities:

If seed = 50 and threshold difference =[15, identify0 2] pixels belonging to the region.

ANS: **Region-Based Segmentation**

(a) Region Growing:
Region growing is a segmentation method that starts from a seed pixel and adds neighboring pixels whose intensity difference is within a given threshold.

(b) Calculation:

Given pixel intensities:

[50, 52, 60, 70]

Seed = **50**

Threshold difference = **15**

Difference from seed:

- 50 → 0 ✓
- 52 → 2 ✓

- 60 → 10 ✓
- 70 → 20 ✗

Answer: Pixels belonging to the region are **50, 52, 60**.

10. Combined Enhancement & Segmentation

(a) Explain why smoothing is applied before segmentation. **(b)** Given image values:

First apply averaging (assume mean =[148) 0, then 5] threshold at $T = 50$.

Determine final segmented output.

ANS: **Combined Enhancement & Segmentation**

(a) Why is smoothing applied before segmentation?

Smoothing removes noise from the image, making object boundaries clearer and improving segmentation accuracy.

(b) Calculation:

Given image values:

[48, 50, 52]

After averaging (mean = **50**):

[50, 50, 50]

Apply threshold **T = 50**:

- 50 → 1
- 50 → 1
- 50 → 1

Answer: Final segmented output = **[1, 1, 1]**

Code of Conduct:

I AMALRAJ P (192524089) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplar y)**	**Level 3 (Proficient)**	**Level 2 (Developin g)**	**Level 1 (Beginning)**
Problem Understan ding	**20%**	**Clearly interprets problem, identifies all parameter s and requireme nts accurately**	**Understan ds problem with minor gaps**	**Partial understan ding; misses key elements**	**Misinterpr ets or unable to understan d problem**

Method Selection	**20%**	**Selects most appropriat e method/for mula with strong justificatio n**	**Selects correct method with minor errors**	**Inappropri ate or partially correct method**	**Unable to select method**
Solution Process	**25%**	**Logical, systematic steps with correct approach throughout**	**Mostly correct steps with minor mistakes**	**Disorganiz ed steps; multiple errors**	**No clear process**
Accuracy of Calculatio n	**20%**	**All calculation s accurate; correct final answer**	**Minor calculation errors**	**Frequent errors; incorrect final answer**	**No valid calculation s**
Interpretati on of Results	**15%**	**Clearly interprets results with units and engineerin g relevance**	**Basic interpretati on with minor omissions**	**Limited or unclear interpretati on**	**No interpretati on**